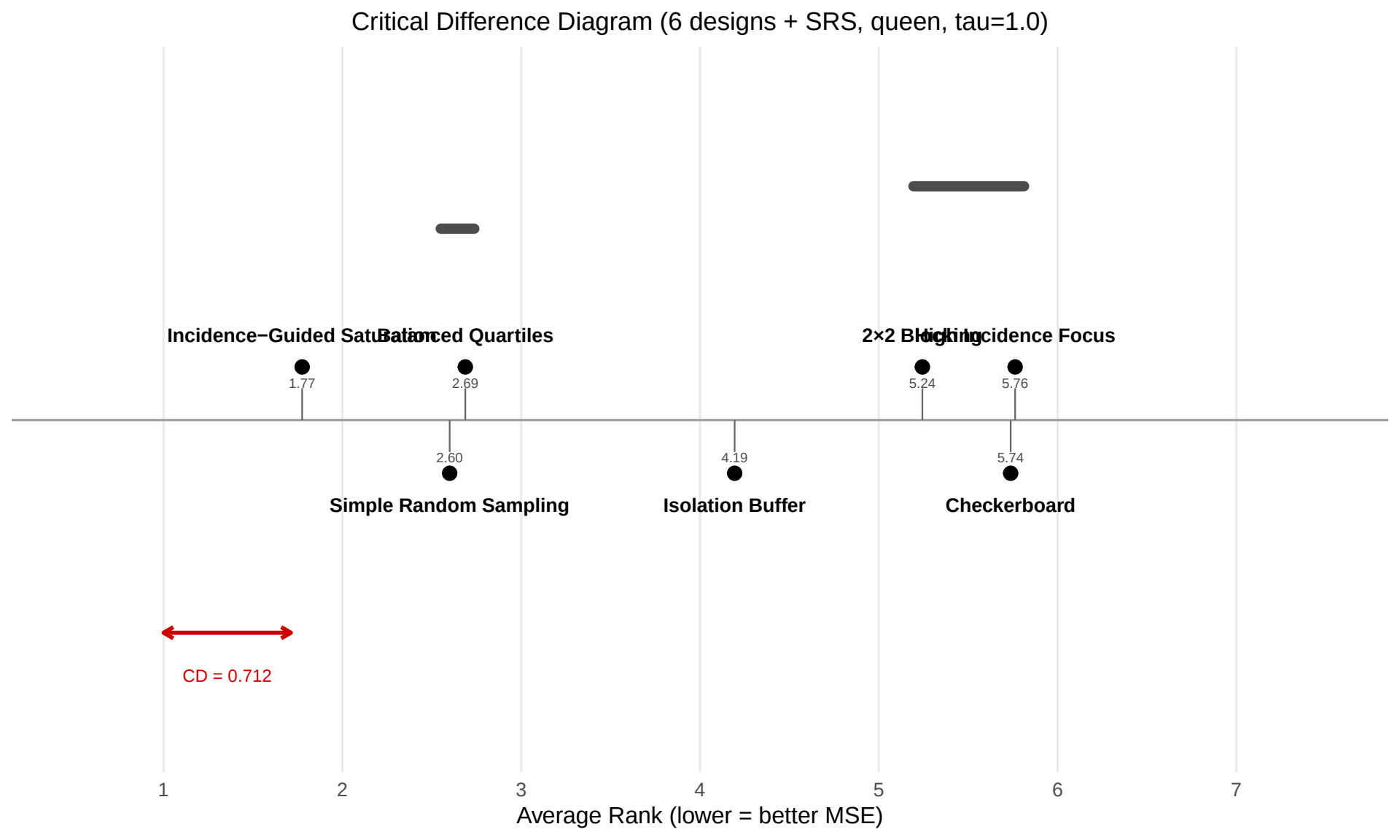
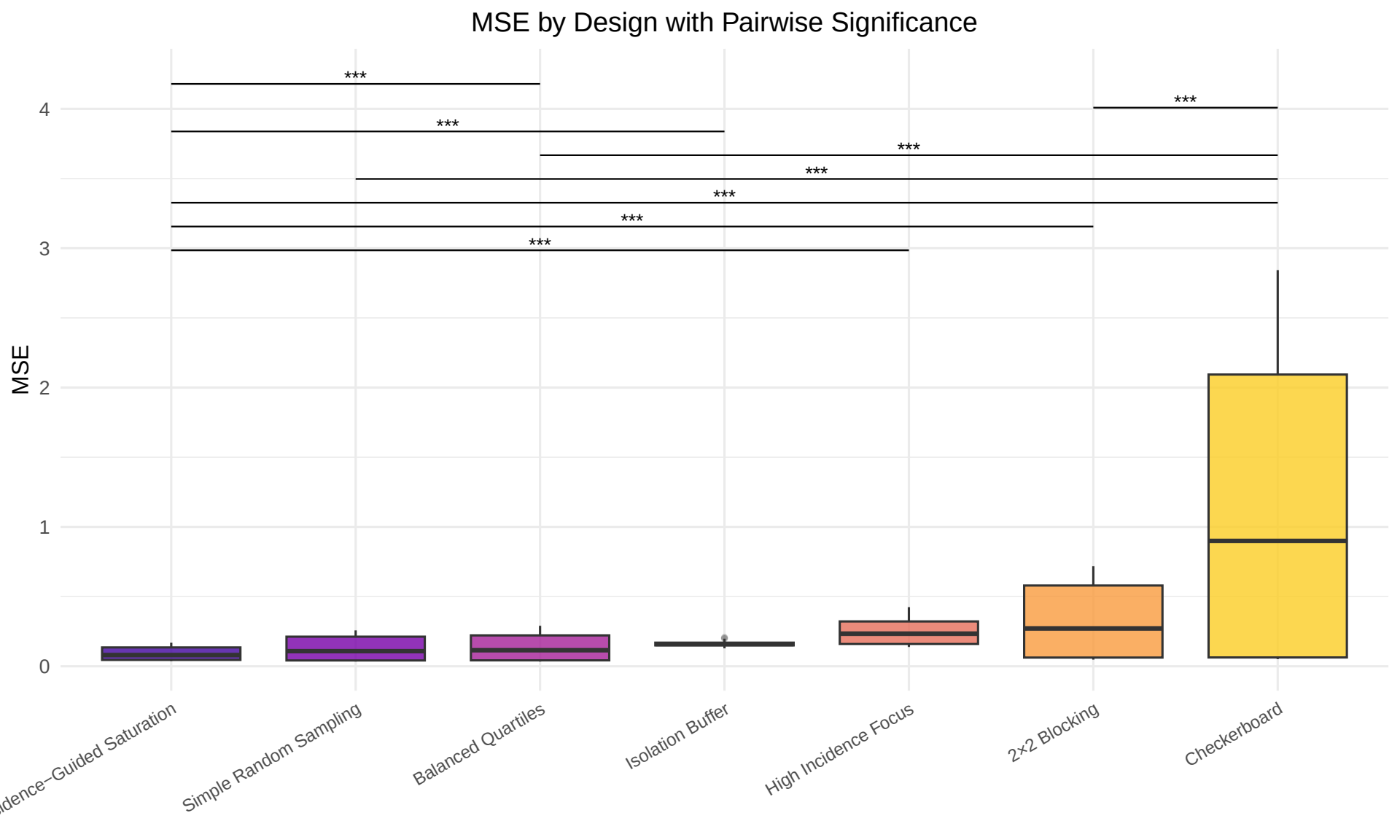
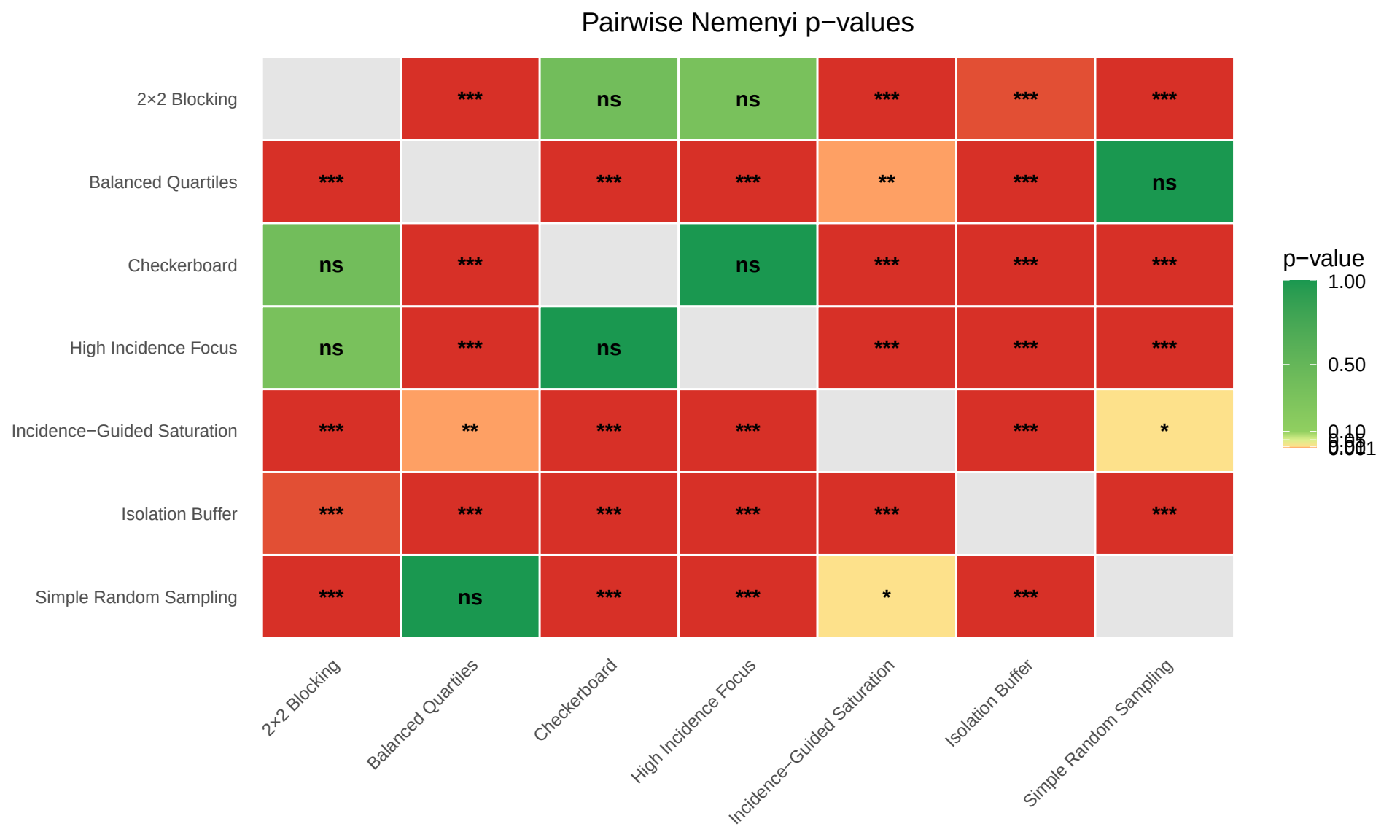
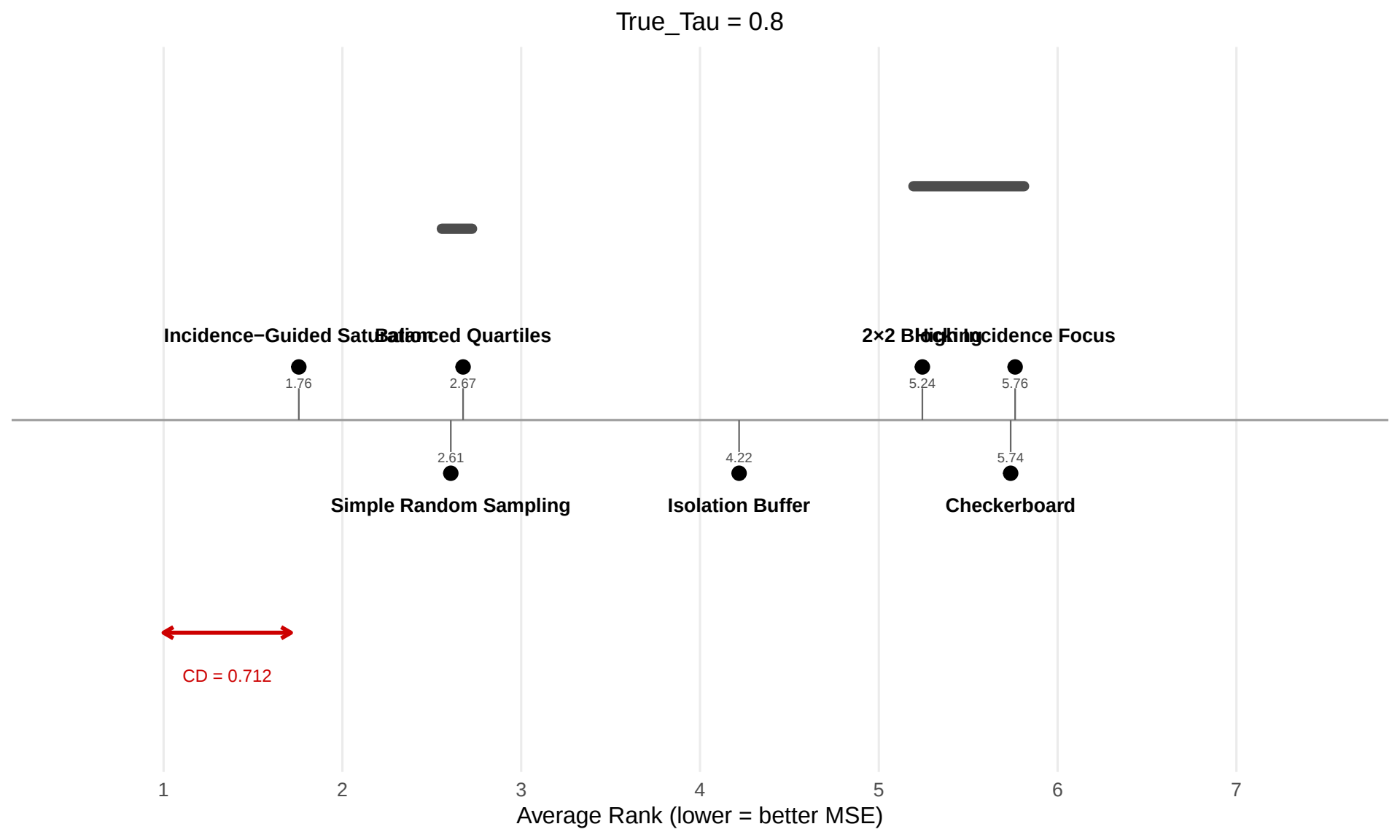
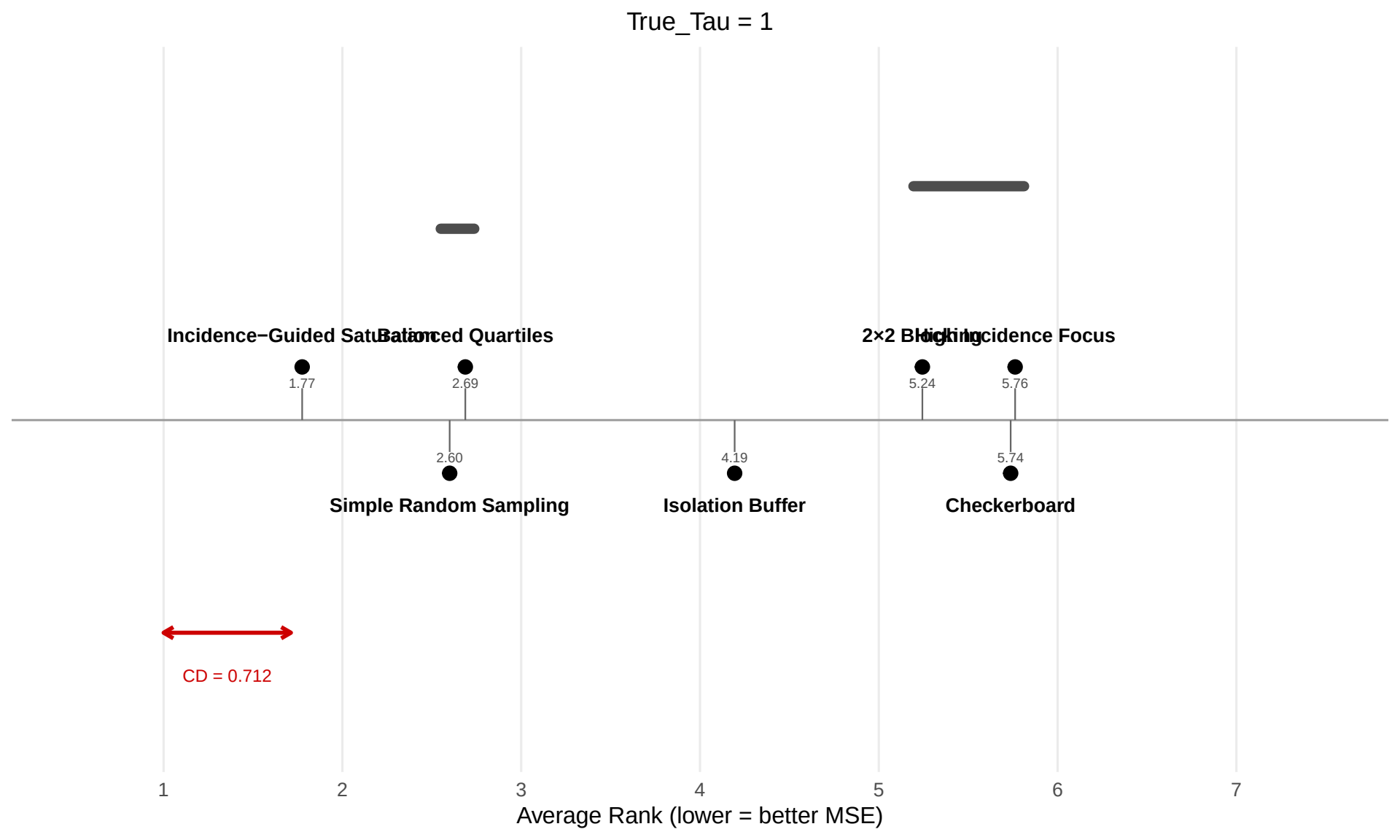

Critical Difference Diagram (6 designs + SRS, queen, tau=1.0)
Incidence−Guided Saturation
1.77
Balanced Quartiles
2.69
2×2 Blocking
5.24
High-Incidence Focus
5.76
Simple Random Sampling
2.60
Isolation Buffer
4.19
Checkerboard
5.74
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)

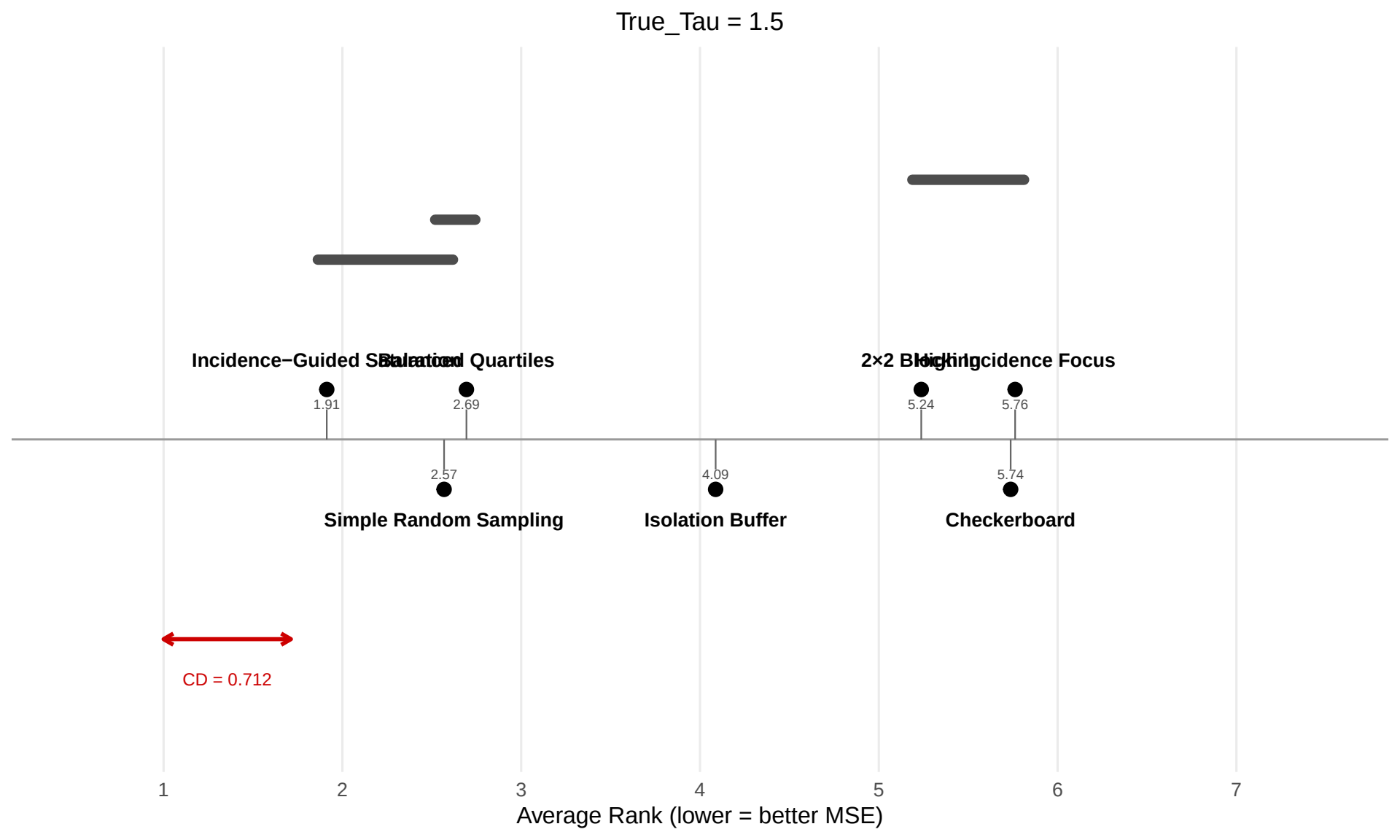

MSE by Design with Pairwise Significance
MSE
0
1
2
3
4
***
***
***
***
***
***
***
idence−Guided Saturation
Simple Random Sampling
Balanced Quartiles
Isolation Buffer
High Incidence Focus
2×2 Blocking
Checkerboard

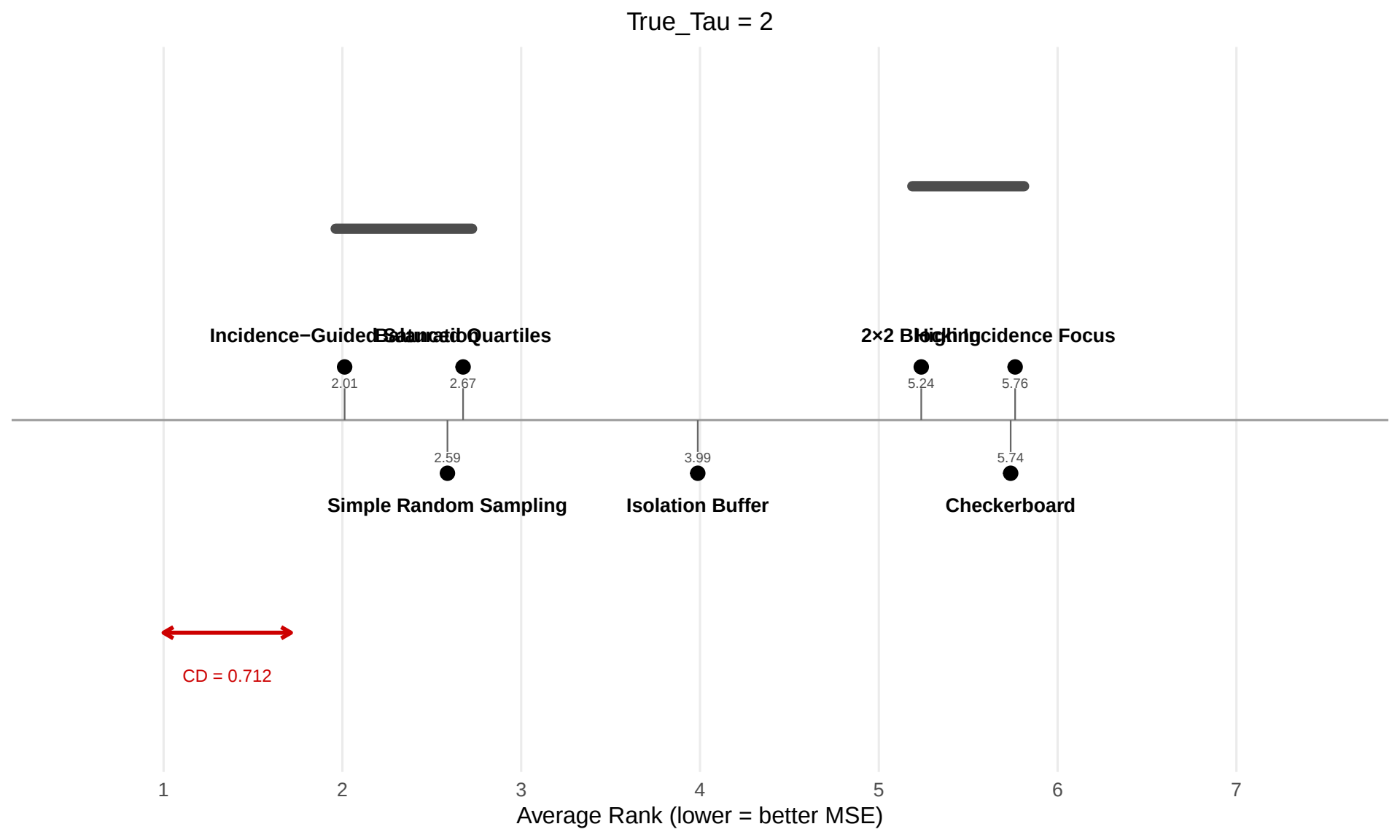
Pairwise Wilcoxon (Holm) p−values (Holm−adjusted)
2×2 Blocking
Balanced Quartiles
Checkerboard
High Incidence Focus
Incidence−Guided Saturation
Isolation Buffer
Simple Random Sampling
***
ns
p−value
1.00
0.50
0.10
0.001

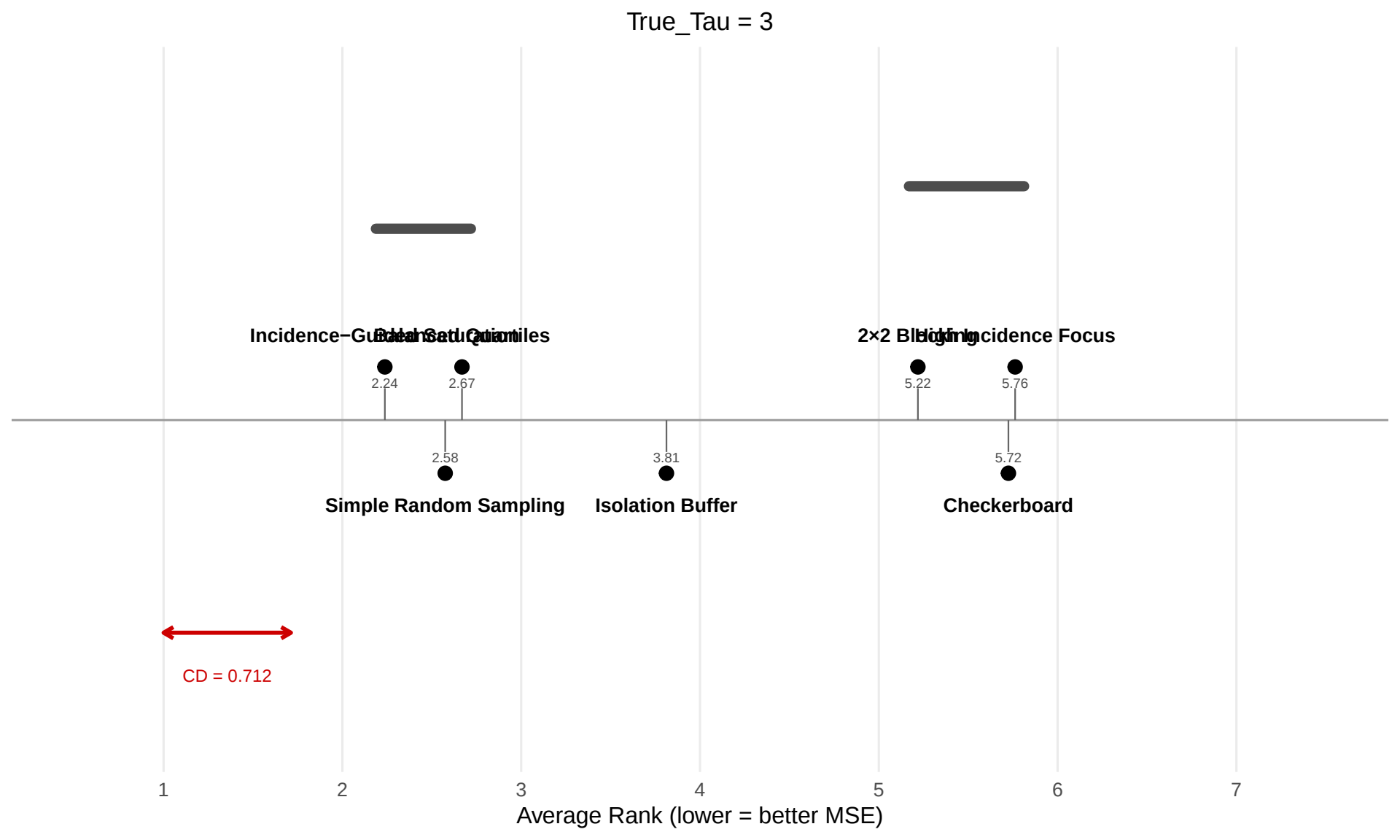

Pairwise Nemenyi p−values
2×2 Blocking
Balanced Quartiles
Checkerboard
High Incidence Focus
Incidence−Guided Saturation
Isolation Buffer
Simple Random Sampling
***
ns
**
*
p−value
1.00
0.50
0.10
0.001


True_Tau = 0.8
Incidence−Guided Saturation
Balanced Quartiles
2×2 Blocking
High Incidence Focus
1.76
2.67
5.24
5.76
2.61
4.22
5.74
Simple Random Sampling
Isolation Buffer
Checkerboard
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)


True_Tau = 1
Incidence−Guided Saturation
1.77
Balanced Quartiles
2.69
2×2 Blocking
5.24
High Incidence Focus
5.76
2.60
Simple Random Sampling
4.19
Isolation Buffer
5.74
Checkerboard
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)


True_Tau = 1.5
Incidence−Guided Stratified
Balanced Quartiles
2×2 Blocking
High-Incidence Focus
1.91
2.69
5.24
5.76
2.57
4.09
5.74
Simple Random Sampling
Isolation Buffer
Checkerboard
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)


True_Tau = 2
Incidence−Guided Balanced Quartiles
2×2 Blocking
High-Incidence Focus
2.01
2.67
5.24
5.76
2.59
3.99
5.74
Simple Random Sampling
Isolation Buffer
Checkerboard
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)


True_Tau = 3
Incidence−Guided Quantiles
2.24
2.67
2.58
Simple Random Sampling
3.81
Isolation Buffer
2×2 Block
5.22
5.76
5.72
Checkerboard
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)